

Chenran Xu

Postdoctoral Researcher on Machine Learning Applications for Particle Accelerators

✉ chenran.xu@kit.edu | 🏠 <https://cr-xu.github.io/> | 📄 cr-xu | 🎓 C. Xu

Education

Dr. rer. nat. in Physics

KARLSRUHE INSTITUTE OF TECHNOLOGY

summa cum laude

Thesis: *Automatic Control of Linear Particle Accelerators with Machine Learning Methods*

Karlsruhe, Germany

Sep. 2020 - Nov. 2024

M. Sc. in Physics

KARLSRUHE INSTITUTE OF TECHNOLOGY

Thesis: *Bayesian Optimization of Injection Efficiency at KARA using Gaussian Processes*

Karlsruhe, Germany

Apr. 2018 - Jun. 2020

B. Sc. in Physics

KARLSRUHE INSTITUTE OF TECHNOLOGY

Thesis: *Investigation of the Long Term Stability of the EDELWEISS Muon Veto System*

Karlsruhe, Germany

Oct. 2014 - Apr. 2018

Experience

Karlsruhe Institute of Technology

DOCTORAL RESEARCHER

Karlsruhe, Germany

2020 - Present

- **Deep Reinforcement Learning:** Applied DRL for automatic accelerator control tasks.
- **Bayesian Optimization:** Implemented BO algorithms for various automatic tuning tasks; Conducted experiments at multiple facilities, including the synchrotron KARA, the ARES linac, and the European XFEL.
- **Differentiable simulation:** Developed backward differentiable beam dynamics code Cheetah
- **NN-based Laser Modulation:** Built spatial light modulator setups for laser profile modulation; Trained U-Net based models to improve the modulation quality.
- **Surrogate Modelling:** Trained neural network surrogate models of a linear particle accelerator.

Karlsruhe Institute of Technology

STUDENT RESEARCH ASSISTANT, HIWI

Karlsruhe, Germany

2018 - 2020

- Research data management and storage
- Established metadata scheme

Publications

Bayesian Optimization of the Beam Injection Process into a Storage Ring

Chenran Xu, Tobias Boltz, Akira Mochihashi, Andrea Santamaria Garcia, Marcel Schuh, Anke-Susanne Müller

PHYS. REV. ACCEL. BEAMS 26 (2023). 2023. DOI: [10.1103/PhysRevAccelBeams.26.034601](https://doi.org/10.1103/PhysRevAccelBeams.26.034601)

Bridging the gap between machine learning and particle accelerator physics with high-speed, differentiable simulations

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia

PHYS. REV. ACCEL. BEAMS 27 (2024). 2024. DOI: [10.1103/PhysRevAccelBeams.27.054601](https://doi.org/10.1103/PhysRevAccelBeams.27.054601)

Reinforcement learning-trained optimisers and Bayesian optimisation for online particle accelerator tuning

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia, Oliver Stein, Erik Bründermann, Willi Kuropka, Hannes Dinter, Frank Mayet, Thomas Vinatier, Florian Burkart, Holger Schlarb

SCIENTIFIC REPORTS 14.1 (2024) p. 15733. 2024. DOI: [10.1038/s41598-024-66263-y](https://doi.org/10.1038/s41598-024-66263-y)

Bayesian optimization algorithms for accelerator physics

Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lübsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez, Brahim Mustapha, Verena Kain, Christopher Mayes, Weijian Lin, Simone Maria Liuzzo, Jason St. John, Matthew J. V. Streeter, Remi Lehe, Willie Neiswanger

PHYS. REV. ACCEL. BEAMS 27 (2024). 2024. DOI: [10.1103/PhysRevAccelBeams.27.084801](https://doi.org/10.1103/PhysRevAccelBeams.27.084801)

Deep Meta Reinforcement Learning for Rapid Adaptation In Linear Markov Decision Processes: Applications to CERN's AWAKE Project

Simon Hirlander, Sabrina Pochaba, Lamminger Lukas, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Annika Eichler, Verena Kain

COMBINING, MODELLING AND ANALYZING IMPRECISION, RANDOMNESS AND DEPENDENCE, 2024. DOI: [10.1007/978-3-031-65993-5_21](https://doi.org/10.1007/978-3-031-65993-5_21)

Towards few-shot reinforcement learning in particle accelerator control

Simon Hirlander, Lukas Lammingier, Sabrina Pochaba, Jan Kaiser, Chenran Xu, Andrea Santamaria Garcia, Luca Scomparin, Verena Kain

- Proc. IPAC'24*, 2024. doi: 10.18429/JACoW-IPAC2024-TUPS60
- Preliminary results on the reinforcement learning-based control of the microbunching instability
L. Scomparin, A. Santamaria Garcia, A. Kopmann, A. Mueller, C. Xu, E. Blomley, E. Bründermann, J. Steinmann, J. Becker, M. Schuh, M. Caselle, T. Dritschler, A. Mochihashi, M. Weber
Proc. IPAC'24, 2024. doi: 10.18429/JACoW-IPAC2024-TUPS61
- Bayesian Optimization for SASE Tuning at the European XFEL
C. Xu, E. Bründermann, A.-S. Müller, A. Santamaria Garcia, S. Tomin
Proc. IPAC'23, 2023. doi: 10.18429/JACoW-IPAC2023-THPL028
- Beam trajectory control with lattice-agnostic reinforcement learning
C. Xu, E. Bründermann, A.-S. Müller, A. Santamaria Garcia, J. Kaiser, A. Eichler
Proc. IPAC'23, 2023. doi: 10.18429/JACoW-IPAC2023-THPL029
- Integration of an Optimizer Framework into the Control System at KARA
C. Xu, E. Blomley, A.-S. Müller, A. Santamaria Garcia, M. Zhang
Proc. ICALEPCS'23, 2024. doi: 10.18429/JACoW-ICALEPCS2023-TUPDP030
- How Can Machine Learning Help Future Light Sources?
A. Santamaria Garcia, E. Bründermann, M. Caselle, G. De Carne, A.-S. Müller, L. Scomparin, C. Xu
Proc. FLS'23, 2024. doi: 10.18429/JACoW-FLS2023-TH3D3
- Optimization Studies of Simulated THz Radiation at FLUTE
C. Xu, E. Bründermann, A.-S. Müller, A. Santamaria Garcia, M. Schwarz, J. Schäfer
Proc. IPAC'22, 2022. doi: 10.18429/JACoW-IPAC2022-WEPOMS023
- Surrogate Modelling of the FLUTE Low-Energy Section
C. Xu, E. Bründermann, A.-S. Müller, A. Santamaria Garcia, J. Schäfer
Proc. IPAC'22, 2022. doi: 10.18429/JACoW-IPAC2022-TUOPT070
- Transverse and Longitudinal Modulation of Photoinjection Pulses at FLUTE
M. Nabinger, A.-S. Müller, M.J. Nasse, C. Sax, J. Schäfer, C. Widmann, C. Xu
Proc. IPAC'22, 2022. doi: 10.18429/JACoW-IPAC2022-TUOPT068
- Machine Learning Based Spatial Light Modulator Control for the Photoinjector Laser at FLUTE
C. Xu, E. Bründermann, A. Eichler, A.-S. Müller, M.J. Nasse, A. Santamaria Garcia, C. Sax, C. Widmann
Proc. IPAC'21, 2021. doi: 10.18429/JACoW-IPAC2021-WEPAB289
- First Steps Toward an Autonomous Accelerator, a Common Project Between DESY and KIT
A. Eichler, E. Bründermann, F. Burkart, J. Kaiser, W. Kuroпка, A. Santamaria Garcia, O. Stein, C. Xu
Proc. IPAC'21, 2021. doi: 10.18429/JACoW-IPAC2021-TUPAB298

Selected Talks and Posters

- | | |
|--|-----------------------|
| 4th ICFA Beam Dynamics Mini-Workshop on Machine Learning Applications for Particle Accelerators | Gyeongju, South Korea |
| TALK & TUTORIAL: <i>APPLYING REINFORCEMENT LEARNING TO PARTICLE ACCELERATORS</i> | 2024 |
| 2nd Collaboration Workshop on Reinforcement Learning for Autonomous Accelerators (RL4AA'24) | Salzburg, Austria |
| TUTORIAL: <i>TUTORIAL ON META-REINFORCEMENT LEARNING AND GP-MPC</i> | 2024 |
| 11th MT-ARD ST3 Annual Meeting | Dresden, Germany |
| TALK: <i>REINFORCEMENT LEARNING APPLICATIONS AT PARTICLE ACCELERATORS</i> | 2023 |
| DPG Spring Meeting of the Matter and Cosmos Section | Dresden, Germany |
| TALK: <i>BEAM TRAJECTORY CONTROL WITH LATTICE-AGNOSTIC REINFORCEMENT LEARNING</i> | 2023 |
| 3rd ICFA Beam Dynamics Mini-Workshop on Machine Learning Applications for Particle Accelerators | Chicago, US |
| POSTER: <i>FAST THZ RADIATION OPTIMIZATION FOR LINEAR ACCELERATOR USING SURROGATE MODEL</i> | 2022 |
| 10th MT ARD ST3 Meeting | Berlin, Germany |
| SPEED TALK & POSTER: <i>BAYESIAN OPTIMIZATION OF SIMULATED THZ RADIATION AT FLUTE</i> | 2022 |
| DPG Spring Meeting Mainz22 | Mainz, Germany |
| TALK: <i>OPTIMIZATION STUDIES OF SIMULATED THZ RADIATION AT FLUTE</i> | 2022 |
| 9th MT ARD ST3 Meeting | Hamburg, Germany |
| INVITED TALK: <i>TOWARDS AUTOMATIC TUNING AND CONTROL OF ACCELERATORS</i> | 2021 |
| DPG Spring Meeting AKBP | Dortmund, Germany |
| TALK: <i>BAYESIAN OPTIMIZATION OF INJECTION EFFICIENCY AT KARA USING GAUSSIAN PROCESSES</i> | 2021 |

Teaching

KIT: Analysis I-II, Further Mathematics I-III

TEACHING ASSISTANT 2016-2018

KIT: Classical Experimental Physics I

TEACHING ASSISTANT 2021

TU Darmstadt: Numerical Methods of Accelerator Physics

GUEST LECTURE: BAYESIAN OPTIMIZATION APPLIED AT PARTICLE ACCELERATORS 2023-2024

Skills

Technical Skills	Python, C++, Matlab, Latex, Git, Unix/Linux, Slurm
DevOps	Docker, Apptainer
Accelerator Physics software	OCELOT, ASTRA, Bmad, Accelerator Toolbox, EPICS
Languages	Chinese (<i>native</i>)
	English, German (<i>professional</i>)